

Sara Logsdon

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Education

University of Pennsylvania <i>Doctor of Philosophy (PhD) Mathematics</i>	Philadelphia, PA Expected 2030
University of Georgia, Morehead Honors College <i>Combined Master of Arts (MA) and Bachelor of Science (BS) in Mathematics</i> <i>Master's Thesis <u>Polynomial Functions over Finite Rings</u> (Supervised by Dr. Pete Clark)</i> <i>Summa Cum Laude</i>	Athens, GA 2025
Budapest Semesters in Mathematics <i>Mathematics courses with honors</i>	Budapest, HU January 2024–May 2024
University of Oxford, Maymester, Keble College <i>Philosophy of Language</i>	Oxford, UK May 2022

Research Experience

NSF REU: Non-Commutative Algebra from Geometry <i>Under supervision of Dr. Hülya Argüz</i>	Athens, GA August 2024
- Mastered non-commutative algebraic techniques with focus on scattering diagrams and deformation quantization - Presented on creating new algebras at the University of Georgia's Algebraic Geometry Seminar	
Hypergraphic Degree Sequence Research <i>Under supervision of Dr. István Miklós</i>	Budapest, HU January 2024–May 2024
- Wrote and proved novel dichotomy theorem characterizing always-graphic 3-uniform hypergraphic degree sequences - Identified new classes of always-graphic k-uniform degree sequences, advancing extremal hypergraph theory	
Quinn Research Group Cilia Project <i>Under supervision of Dr. Shannon Quinn</i>	Athens, GA January 2022–December 2023
- Constructed weakly-supervised segmentation methods for ciliary motion detection in biomedical videos, leveraging variational autoencoders for manifold learning - Designed original recurrent neural network regularization technique that resolves vanishing gradient problem through use of unitary matrices	
University of Georgia Mathematics Directed Reading Program <i>Under supervision of Dr. Jimmy Dillies</i>	Athens, GA August 2022–December 2022
- Mastered advanced concepts in projective geometry and representation theory, studying Kirillov's "What are Numbers?" - Independently authored a paper developing foundational theory for elliptic curve cryptography through the construction of group structures on conic sections	
Research on Topological Polymers <i>Under supervision of Dr. Jason Cantarella</i>	Athens, GA September 2021–September 2022
Created computational methods to analyze theta curves and line arrangements using Wolfram Mathematica	

Work Experience

Johns Hopkins Applied Physics Laboratory <i>Combat Systems and Analysis Intern</i>	Laurel, MD May 2023–August 2023, May 2024–August 2024
- Implemented network topology algorithms in MATLAB and C++ to a United States Navy project, enabling efficient communications between ships and planes - Wrote original single and multi-beam scheduling algorithms for the U.S. Navy using linear optimization in Python	
Department of Energy Idaho National Laboratory <i>Cybersecurity Intern</i>	Idaho Falls, ID June 2022–December 2023
- Analyzed formal methods applied to cyber-physical models of critical infrastructure and the cyber-physical critical security and resilience properties of such models and infrastructure - Developed a formal methods-based approach to security using homomorphic encryption and theorem provers	
Star Tutors <i>Mathematics Tutor</i>	Remote September 2019–May 2025
Tutored high school students in algebra, calculus and statistics	

Honors and Awards

National Defense Science and Engineering Graduate (NDSEG) Fellowship (2025-2028)

National Science Foundation (NSF) Graduate Research Fellowship Program, Honorable Mention (2025)

Joy P. Williams Physical Sciences Award (2025)

- Selected by the Morehead Honors College; awarded to an outstanding female student studying physical sciences

Barry M. Goldwater Scholarship (2024)

Charles M. Strahan Award (2024)

- Selected by University of Georgia mathematics faculty; first place among all third-year mathematics majors

University of Georgia Libraries Undergraduate Research Award, 1st Prize (2024)

- Awarded to my paper “On the Group Structure of Conic Sections and Elliptic Curves” for excellence in student inquiry

University of Georgia Center for Undergraduate Research Award (Summer 2022, Fall 2022, Spring 2023)

- Awarded by University of Georgia’s Center for Undergraduate Research Organization to top undergraduate researchers

Hollingsworth Award (Fall 2021, Spring 2022)

- Top student in select early upper division math courses (awarded for MATH 3500 and MATH 3510)

University of Georgia Foundation Fellowship (2021–2025)

- University of Georgia’s premier undergraduate merit-based scholarship, covering full cost of attendance, research and conference grants, and spring travel-study

Publications and Presentations

Publications

- S. Logsdon, *Unlocking Sudoku’s Secrets*. Chalkdust Magazine, Issue 21. <https://chalkdustmagazine.com/features/unlocking-sudokus-secrets> (2025).
- S. Logsdon, *Exploring Applied Cryptosystems to Formally Verify Security in Cyber-Physical Systems*. U.S. Department of Energy Office of Scientific and Technical Information. <https://www.osti.gov/biblio/1880065> (2022).

Submitted

- S. Logsdon, *On the Group Structure of Conic Sections and Elliptic Curves*. Under review for publication at: Rose-Hulman Undergraduate Mathematics Journal.

Preprints

- S. Logsdon, I. Miklós, A. Maheshwari, A. Zhang, *A Dichotomy Theorem on the Complexity of 3-Uniform Hypergraphic Degree Sequence Graphicality*. <https://arxiv.org/abs/2411.19049>. (2024)
- S. Logsdon, *Fully Homomorphic Encryption: A Mathematical Introduction*. <https://eprint.iacr.org/2023/1402> (2023).

Presentations

- S. Logsdon, A. Zhang, *A Dichotomy Theorem on the Complexity of 3-Uniform Hypergraphic Degree Sequence Graphicality*. Presented at: 2025 Joint Mathematics Meeting, Seattle, WA, January 2025.
- S. Logsdon, *Deformation Quantization for Creating Non-Commutative Algebras*. Presented at: University of Georgia Algebraic Geometry Seminar, Athens, GA, October 2024.
- S. Logsdon, D. Ott, V. Schneider, *Stochastic Unitary Constraints*. Presented at: SciPy Scientific Computing with Python, Austin, TX, July 2023.
- S. Logsdon, *Group Theory and Geometry*. Presented at: University of Georgia Mathematics Directed Reading Symposium, Athens, GA, December 2022.

Extracurricular and Community Service

University of Georgia Women’s Club Ultimate Frisbee

August 2022–May 2025

The Backpack Project of Athens, Director of Donations and Partnerships

August 2022–December 2023

Pre-College Research Institute, Director of Math and Physics

September 2021–September 2022

Skills

Java, Python, MATLAB, Wolfram Mathematica, Git, Linux, L^AT_EX